



DISCRETE MATH

Discrete mathematics is the study of mathematical structures that are fundamentally discrete rather than continuous.

In contrast to real numbers that have the property of varying "smoothly", the objects studied in discrete mathematics do not vary smoothly in this way, but have distinct, separated values.

Discrete mathematics therefore excludes topics in "continuous mathematics" such as calculus and analysis.

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The study of how discrete objects combine with one another and the probabilities of various outcomes is known as **combinatorics**.

What is a Set theory?

Set Theory is the mathematical theory of well-determined collections called **sets**. A **set** is an unordered collection of objects. The term object has been used without specifying what an object is. The 'objects' that belongs to a given set is called an **element** or **members** of that set.

NOTATIONS

Curly brackets {...} are used to stand for the phrase 'the set of ...'. The curly brackets are sometimes called "set brackets" or "braces". These braces can be used in various ways.

{2, 4, 6, 88, ...}

Elements or members

Three dots or ellipsis which means continue on (infinity)

Curly brackets can be used in various ways.

- We may list the elements of a set:
{-2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8}
- We may describe the elements of a set:
{integers between -2 and 8, inclusive}
- We may use an identifier (the letter 'x' for example) to represent a typical element, a symbol to stand for the phrase 'such that', and then the rule or rules that the identifier must obey:
- Set comprehension notation. The symbol $\in$ is used as follows:
 $\in$ means 'is an element of...'
 Blossom $\in$ {Powerpuff Girls}
 $\notin$ means 'is not an element of...'
 Allan $\notin$ {Snow White's seven dwarfs}



A set can be finite or infinite.

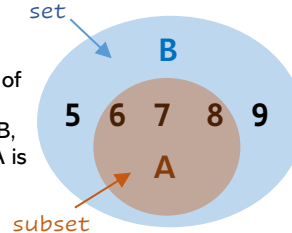
{ball, ring, bell, flower, ...} $\rightarrow$ infinite set
 {1, 3, 5, 8} and {a, b, c, ..., x, y, z} $\rightarrow$ finite set

Universal Set

Universal set is the set of all the 'objects' currently under discussion. The universal set doesn't contain everything in the whole universe. On the contrary, it restricts us to just those things that are relevant at a particular time. It is denoted by (bold letter) **U**.

Subsets

If all the elements of a set A are also elements of a set B, then we say that A is a **subset** of B, and we write: $A \subseteq B$



In other words, A is a subset of B if and only if every element of A is in B, like that in the figure above.

Power Subsets

$A \subseteq B$ does not imply that B must necessarily contain extra elements that are not in A. If B does contain at least one element that isn't in A, then we say that A is a **proper subset** of B. In such case, we would write: $A \subset B$

A is a proper subset of B if and only if every element of A is also in B, and there exists **at least one element** in B that is **not** in A.

Examples:

{1, 2, 3} is a subset of {1, 2, 3}, but is NOT a proper subset of {1, 2, 3}.

{1, 2, 3} is a proper subset of {1, 2, 3, 4} because the element 4 is not in the first set.

Note that when A is a proper subset of B, then it is also a subset of B.

Empty (or Null) Set

Empty or null set is the set containing no elements at all. It is denoted by a pair of empty braces {} or by the symbol $\emptyset$.

Example: If $U = \{\text{whole numbers}\}$ the

$$\{x | x^2 = 10\} = \emptyset$$

Note: The empty set is a subset of every set, including the empty set itself.

Order

Order in sets does not mean the order of the elements but the size of the set. In sets, it doesn't matter what order the elements are in. Another term for order is **cardinality**.

{1, 2, 3, 4} is the same set as {3, 2, 4, 1}

Superset

A set A is a **superset** of another set B if all elements of the set B are elements of the set A. The superset relationship is denoted as $A \supset B$.

Example:

A {cat, dog, hen, bird} and B {cat, dog, hen}, then $A \supset B$ but $B \not\supset A$. Since A contains elements not in B, the A is a **proper superset** of B.

Disjoint Sets

Two sets A and B are called **disjoint** if A and B have no elements in common. Another equivalent definition of **disjoint sets** would be that the intersection of the two sets actually equals the empty set.

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Examples of disjoint sets:

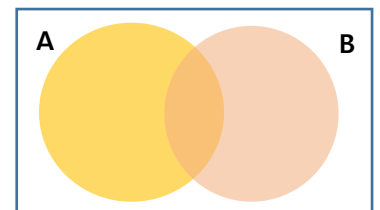
- {1, 2, 3} and {a, b, c}
- {Carlo, Enzo, Judd} and {Kaye, Lynn, Honey}

Venn Diagram

A Venn diagram uses overlapping circles or other shapes to illustrate the logical relationships between two or more sets of items. Venn diagrams were conceived around 1880 by **John Venn**.

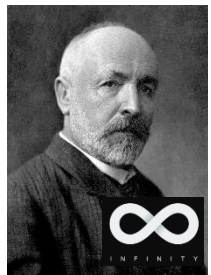
The universal set is represented by a rectangle.

Points inside the rectangle represent elements that are in the universal set. Points outside represent things not in the universal set.



Until the end of the nineteenth century no mathematician had managed to describe the **infinite**, beyond the idea that it is an absolutely unattainable value.

Georg Cantor was the first to fully address such an abstract concept, and he did it by developing set theory, which led him to the surprising conclusion that there are infinities of different sizes.



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